

Topic 1 Review [100 marks]

1. [Maximum mark: 4]

Describe **two** advantages of using surveys to determine user requirements for updating a computer system.

[4]

Markscheme

*Award **max** [4].*

*Award [1] for an advantage and [1] for a description, **x2**.*

Fast/cost effective/flexible way of gathering data;
from large number of users/surveys can be sent out as offline or online form
(for better reach);

Standardized question formats;
can yield more reliable data than other methods (interviews, observation,
etc.);

Anonymity/privacy for respondents;
can lead to more honest responses;

Simple to administer (less time consuming than other methods);
A large amount of data gathered in a short period of time;

Fast results/easy to extract quantitative data/easy to analyse data;
because all the users are asked the same questions;

Note: *Reward other correct responses.*

Examiners report

A small number of students managed to earn all four marks for this question. Quite a few students simply assumed that surveys can only be

online and went on to describe the advantages of having it online which helped them gain some partial marks.

2. [Maximum mark: 2]

Identify **two** methods of implementing a new computer system.

[2]

Markscheme

*Award **max** [2].*

Parallel running;
Pilot running;
Direct changeover;
Phased conversion;

Examiners report

Most students were able to achieve full two marks here.

3. [Maximum mark: 2]

Describe the role of a DNS server.

[2]

Markscheme

*Award **max** [2].*

contains a database of public IP addresses and their associated hostnames/domain names;

translates/matches those website hostnames to their corresponding IP addresses;

translates human-readable domain names (like www.ibschool.com) into machine-readable IP addresses (like 2001:0db8:85a3:1111:0000:8a2e:0380);
allows users to access websites by using meaningful names (instead of IP addresses which are difficult to remember)/presents the corresponding web page of the domain name entered by the user;

acts like a directory of web servers on the internet;
lets web browsers connect to the web servers;

converts human-readable domain names into IP addresses;
ensures that internet traffic reaches the correct destination based on the domain name;

Note: *Reward other correct responses.*

Examiners report

SL:

Some students found the role of DNS server challenging and gave vague responses which were mainly guess work.

HL:

Many students wrote their answers relying on guesswork and luck. It is evident that these students do not know what the role of a DNS server is.

4. [Maximum mark: 15]

An architect uses a single computer for her business. She uses hand-drawn sketches to present her ideas to clients and computer-aided design (CAD) software to create construction projects, alterations, and redevelopments.

(a.i) State what is meant by CAD software.

[1]

Markscheme

*Award **max** [1].*

software used (by designers) to create 2D/3D models of physical components/software used to simulate hand drawn sketches or component parts of an entity (machine, vehicle, etc.);
software that is used to create/modify/analyse/enhance a design;

Examiners report

Some students responded by providing only the full form of CAD, thus losing a mark. Additionally, some repeated phrases from the question, like 'it is used to create construction projects, alterations' etc. which resulted in losing a mark.

(a.ii) Outline **two** benefits for the architect of using CAD software.

[4]

Markscheme

*Award **max** [4].*

*Award [1] for a benefit and [1] for a reasonable expansion, **x2**.*

Example answers:

Increased productivity;

CAD software automates most of the architect's tasks which allows quicker project completion/the same amount of time can result in a greater number of completed projects;

Flexibility to modify designs;

Allows for changes and revisions to be visualized and approved;

Decreased architect's effort;
the different parts of code and drawings can be modified easily then reused multiple times;

Saves time;
Easy/quick to create/edit/make alterations in the drawings;

Saves cost/money;
a model could be created using CAD, and modified before the real thing is produced (if it does not work, it could cost the architect more money to create again);

Better quality of drawings;
The CAD makes use of best tools (that allow creation of precise complex drawings of surfaces and shapes);

Improved accuracy;
CAD tools include many mathematical equations that perform vital calculations/examine curves/surfaces;

Fewer errors;
the CAD software makes use of tools to investigate an error, diagnose the problem, and solve it;

Easy to share work with colleagues/clients;
The CAD tools make it easier to save the files and store it in a way that they can be used again and/or sent;

Integration with other tools;
CAD designs can be given to a 3D printer or CAM for direct implementation;

Note: *Reward other reasonable responses.*

Examiners report

Most students struggled to clearly identify the two benefits. Vague responses received no marks. For instance, simply stating that “it is easy for the client to see the product” cannot earn a mark without a reasonable explanation, such as mentioning simulation or 3D models.

The architect often creates large graphic files that are sent via the internet.

- (b) Explain the need for data compression software in storing and transferring these large graphic files.

[3]

Markscheme

*Award **max** [3].*

To reduce the size of files;
so that less memory space on the architect's computer is occupied/more CAD files can be stored in the available memory;
faster upload/download/transmission time reduced/improved data transfer speed;
easier to meet attachment size restrictions/reductions in communication bandwidth (capacity for the data transfer):
meaning a significant decrease in expenses;

Examiners report

While there were a few good answers to this question, many students need to improve their understanding of what compression is. Responses included vague references to packet switching and prevention of loss of data using compression. Those who were aware of the concept were able to achieve all three marks. Students must understand that simply stating “Compression makes transfer easier” will not earn any marks. They need to

clearly express that “Compression reduces the size of the file, thus taking less time to transfer data.”

Data loss can cause downtime, which can force the architect to pause business operations.

The impact on productivity depends on the amount of data lost as well as the time it takes for data recovery.

(c.i) Describe how the architect could minimize downtime if the disk system on her computer fails.

[3]

Markscheme

*Award **max** [3].*

Award [1] for a reasonable method;

Award [1] for indicating how downtime is minimized;

Award [1] for indicating how data is made available;

Example answers:

A regular/periodical back up;
stored on external storage devices;
data could be quickly recovered

A secondary system/a standby computer (or only duplicate disks and a processor);
with an appropriate backup system;
that might take over with data quickly restored to the last point it was backed up;

A failover system/RAID;
that automatically takes over;
with no data loss;

in case of a hard drive failure not all of data is erased/damaged;
she can (buy and) use a data recovery software/hire a skilled data recovery service;
this will enable recovery of as much data as possible, but it takes time;

Cloud Storage/the architect can save/back up all files to the cloud (*Accept examples, OneDrive, Google Drive, etc.*);
and can access them from any device/anywhere;
anytime/when required/when disk fails;

Examiners report

Most students did not fully grasp the concept of minimizing downtime. Some managed to earn one mark by mentioning backups, a failure system, or cloud solutions, but failed to describe how that could help reduce downtime.

(c.ii) Outline **two** causes of data loss **other than** hardware failure.

[4]

Markscheme

*Award **max** [4].*

*Award [1] for a cause and [1] for a reasonable expansion, **x2**.*

Human errors;

Data may be accidentally deleted/entered/alterred inaccurately/files may be wrongly replaced/files might not be saved;

Transfer errors;

Data may be lost in transit between two systems (such as a network failure, network congestion, incorrect storage destination, between two physical locations – transporting a backup/storage device to another location);

Computer viruses/malicious acts/malware/hacking;

Can steal/alter/destroy data;

Software malfunction;
an application used for requesting/editing data can crash resulting in data loss;

Natural disasters (for example, floods, earthquakes, hurricanes, cyclones, fires, lightning);
that can damage on-premises software/hardware and lead to data loss;

Power failures;
can damage operating system resulting in data loss;

Disk full/no available space on the disk;
prevents new (or updated) data from being saved leading to data loss;

Lossy compression;
compressed file does not restore/rebuilt in its original form;

Note: *Reward other reasonable responses.*

Examiners report

Several varied answers were seen here with many students achieving a full score.

5. [Maximum mark: 15]

An organization is implementing a new computer system.

- (a) Identify **two** organizational issues related to the implementation of the new system.

[2]

Markscheme

Award [2 max]

User roles / the organization restructure their workflow;

Technology issues / issues of software compatibility / hardware compatibility;

Examiners report

Candidates were usually able to identify at least one organizational issue related to user roles or technology, when implementing a new system. In many cases, candidates got both.

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The management considered phased conversion and direct changeover as methods of implementation.

(b) Evaluate these **two** methods of implementation.

[5]

Markscheme

Award [5 max]

Direct changeover:

is the cheapest and quickest/the old system is completely switched for the new one;

this is straight forward but also the most risky / nothing to fall back on;

no need to keep data duplicates;

it allows the organization to change the system when most convenient;

the employees have very little time in order get use to the new system as the change is instantaneous;
there is a period of time when neither systems are operational;

Phased conversion:

method where the old system is still in use but parts of the new system or modules are introduced, involves bringing in the new system one step at a time;
less risky than direct changeover; less risky that the whole system will go wrong/if something happens, it will only affect the specific part;
takes a lot of time;
employees have enough time for training/to get use to the new system / are introduced to the changes in small stages;
Employees/users could ask for changes which then hold up the installation of the next phase which helps improving the system;

Examiners report

The vast majority of candidates were able to correctly evaluate the two given methods of implementation, direct changeover and phased conversion, with very comprehensive contextual responses seen. In some cases, it wasn't quite clear whether the candidate was describing phased conversion or pilot implementation. However, these cases were rare.

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(c.i) State **one** type of testing that involves users. [1]

Markscheme
<i>Award [1 max]</i> User acceptance test; Beta testing;

Examiners report
The vast majority of candidates correctly stated beta testing, as a type of testing that involves users.

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(c.ii) Identify **three** consequences of inadequate testing. [3]

Markscheme

Award [3 max]

Can lead to software which is not appropriate for the purpose it was intended/can lead to the system not meeting user requirements;
Can lead to (undiscovered) bugs in software/errors in the system;
Can lead to end user dissatisfaction;
Can lead to reduced (employee) productivity;
Can lead to decreased reliability of the organization;

Examiners report

Candidates generally demonstrated their understanding of the consequences of inadequate testing with many full or nearly full mark responses seen. Some candidates unfortunately gave good descriptions of the consequences that concentrated on the same marking point, for example, they described the software not meeting user requirements or being fit for the intended purpose in multiple ways, therefore limiting their mark.

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- (d) Discuss the social and ethical issues associated with the introduction of a new computer system.

[4]

Markscheme

Award [4 max]

Personal/professional development of all employees must be considered;

Physical safety (of all users);
Ergonomic standards (human-computer components);
Human dignity of all users;
The new system might be designed to replace some staff;
Code of ethics (system resources should not be used without approval);

Examiners report

Candidates recognised the more obvious social and ethical issues associated with the introduction of a new computer system, such as the need to replace some staff, or issues related to the implementation of a code of ethics. However, many candidates concentrated too much on matters related to poor data protection issues. In addition, the issues related to ergonomics, physical safety at work or professional development were rarely seen.

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6. [Maximum mark: 3]

(a) Identify **one** cause of data loss.

[1]

Markscheme

Award [1 max].

Any malicious activity;

Natural disasters;

Human error;

Examiners report

The vast majority of candidates identified a relevant cause of data loss, such as malicious activity, natural disasters or human error.

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- (b) Describe **one** way offsite storage can be used to prevent data loss.

[2]

Markscheme

Award [2 max].

Internet backup service could automatically back up (all important) data files;

to a remote server that could be accessed/controlled over the internet;

Copies of all important data files (backup) could be placed on two separate hardware devices;

which are placed in two different physical locations;

A remote file server could be set up;

for uploading all important data files;

Examiners report

Candidates were generally able to identify appropriate methods of offsite data storage, but clear descriptions of how these could be used to prevent data loss were less common, with answers simply describing the method rather than how it could prevent data loss.

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7. [Maximum mark: 15]

Website developers need to consider a range of usability factors when designing a website.

- (a) Identify **two** usability factors that need to be considered in the design of a website.

[2]

Markscheme

Award [2 max].

complexity / simplicity / amount of effort to get a result / number of errors with the time taken to move past them;

readability / comprehensibility / reading or writing speed;

learnability / time to accomplish tasks on the first use;

effectiveness (user performance);

efficiency (time needed to complete a task);

Examiners report

Answers that simply identified usability factors related to website design, such as: complexity, readability, learnability, effectiveness and efficiency, were rare. The vast majority of candidates opted for a more descriptive approach, which generally did not reflect the required answers.

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- (b) Outline **one** reason why visual displays on a computer screen can create difficulties for some people.

[2]

Markscheme

Award [2 max].

Award marks for the reason/accessibility issue arising from visual / hearing impairments / cognitive disabilities, and not only for users with disabilities but also for those using mobile devices (alternative browsing devices such as TVs, watches, etc.), or those with slow network connections.

Visual display design is not logical;
workflows are not simple, and do not require as few interactions as possible to complete;

Visual display design is not consistent;
navigation, header, footer, and main content are not always in the same places;

Visual display design is not usable as possible;
tools are not easy to use, processes are not broken down into logical steps;

People with poor eye-sight/color blind can have difficulties to distinguish; because of the way graphics, words and directions are used / combinations of some colours (e.g., red and green);

Examiners report

Candidates in general offered responses to this question related to visually impaired users, and some gained marks here, particularly if their response related to poor choice of colours making it hard to read, for example. However, candidates could have also given answers related to screen layout not being logical, consistent screen layout design, or the fact that consideration needs to be given to the size of the screen being used, such as on mobile devices.

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A company promotes its products online. To make a purchase, customers are required to register with the company and provide data like their name, date of birth, age, gender and email address. Once registered, more than one customer is able to access the server to retrieve and modify their data at the same time.

- (c.i) State where the customer data is held during the process of modifying their data.

[1]

Markscheme

Award [1 max].
RAM/primary memory;

It's held by the process/thread handling the customer access, which may be in RAM or in other storage if the process is swapped out while handling the access;

Examiners report

Some candidates recognised that customer data is held in RAM while it is being modified. However, other candidates incorrectly stated that it would be held in the computer or the server, which is not specific enough to achieve the mark.

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- (c.ii) Explain how the operating system ensures that each customer's data is secure when multiple users are accessing the data at the same time.

[4]

Markscheme

Award [4 max].

OS (memory management function) allocates / deallocates memory to each process/customer task, and guarantee each customer task the resources it needs to run correctly;
moves processes back and forth between main memory and secondary memory during execution to prevent overwriting / accidental interchange;
OS (hardware memory protection- part of an OS) isolates/protects customer's data/applications;
and control access rights to the specified memory area (for example, prevents write access to the memory which is not allocated to the process/customer task);
and protects data / applications when in memory / RAM from malicious

code (prevents attempts to execute the contents of the partition/ allocated memory);

Examiners report

The majority of candidates did not recognise that this question related to how an operating system functions and keeps data secure in multiuser environments. Candidates offered answers related to security, but often in relation to authentication when logging in. Some marks were achieved for statements related to the operating system's role in preventing one user's data overwriting that of another, or for the operating system controlling access rights to different parts of the memory. However, high marks on this question were rare.

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- (d) The company is considering sharing its customers' data with marketing organizations.

Explain why there could be ethical issues for the company when sharing its customers' data.

[6]

Markscheme

Award [6 max].

privacy of customer/ person data;

company needs to gain permission from customer;

customer would need to be able to view all data details;

customer must be informed about all uses that will be made of data;

customer must be informed to whom data will be disclosed.

legal issues to do with unauthorized disclosure of customer data;

Examiners report

Candidates had the opportunity to discuss ethical issues related to a company considering sharing customers' data without permission. Many candidates described one or more of the points: privacy of its customer data, the need to obtain permission from customers in order to share their data, and the legal issues that may arise due to the unauthorised disclosure of customer data. However, candidates could also have discussed the need for customers to be aware of how their data may be used, the actual data that may be being shared and to whom it would be disclosed. Most candidates achieved some marks here, but mostly in the centre of the mark range.

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8. [Maximum mark: 15]

A school has a local area network (LAN) connecting its computers and peripheral devices. The LAN also provides access to the internet.

(a) Describe the role of a router in this network.

[3]

Markscheme

Award [3 max].

controls the flow of data in the network;

inspects address of data packets;

directs to the appropriate network path/ selects a path between networks
(by inspecting address of data packets);

securely transmits data packets (across that path toward the intended
destination);

Examiners report

Candidates generally recognised that a router directs data between networks and gained a mark for this. However, most answers didn't really go beyond that point and concentrated more on access to the internet. Some candidates did recognise the role of addresses in data packets to ensure they arrived at the correct destination.

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Users have been troubled by slow speeds when accessing the internet.

- (b) Outline **two** reasons why there might be a reduction in data transmission speed at certain times.

[4]

Markscheme

Award [4 max].

malicious software/spyware/malware / viruses;
attacking the system;

the type of connection;
valid outline for the slow down (depending on the type of connection);

Note: *There are many types of internet connection: wireless connections, wireless hotspots, mobile internet, broadband connections, fixed wireless and satellite connection, cable, DSL;*

hardware malfunction;
internet speed can considerably slow down if modem-router configuration
is not correct / updated;

external factors;
during the peak hours of daytime the most visited websites encounter
more network traffic than what has been anticipated;

NOTE: *Accept examples. For example, too many students try to connect to an e-learning site at the same time.*

Mark as 2 and 2

Examiners report

The vast majority of candidates who achieved marks in this question, which asked for two reasons why there may be a reduction in data transmission speed at certain times, gave answers that may fit into category 'external factors'. An example of this was high demands on bandwidth due to many

students accessing the network at the same time. Sometimes, candidates gave more than one answer related to this same category. Other categories, such as malicious software, type of connection and hardware malfunction, were largely overlooked by candidates.

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- (c) Outline **two** measures that the school could take to safeguard its data from unlawful access via the internet.

[4]

Markscheme

Award [4 max].

Preventing unauthorized access/No access without user names/passwords; which should be regularly changed/made difficult to guess;

Data should be encrypted;

Allowing only those with decrypting code access;

Teachers/students should be trained for safe practices;

to create a risk-conscious /security-aware culture within the school;

Installation of virus checkers/spyware software;

to prevent damage to data files or the system / to prevent data being extracted from the files/system;

Mark as 2 and 2

Note: Accept appropriate measures such as physical security measures / access control / data security / network and communication security / user awareness and education, etc.

Examiners report

Candidates recognised at least one measure that a school could take to safeguard its data from unlawful access via the internet, with correct answers based on authentication of users or encryption. A small number of candidates recognised the benefits of anti-virus software. However, candidates could also have discussed the benefits of training staff or users in safe practices.

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- (d) The inventory of office supplies used in the school is stored on the computer as a single file.

Each of the office supplies in the inventory (such as paper, ink, toner, printers, pens, staplers, pencils and scissors) has a unique ID number, name, maximum quantity, minimum quantity and remaining quantity.

Outline the steps in an algorithm that would output a list of supplies with the quantity to be ordered.

[4]

Markscheme

Award [4 max].

Award [1] for looping through the inventory file;

Award [1] for reading the supply data record (ID number, name, maximum quantity, minimum quantity and remaining quantity);

Award [1] for checking if the quantity remaining is less than minimum quantity;

Award [1] If so, calculate the quantity to be ordered (number to buy) / as the maximum quantity minus the quantity remaining;

Award [1] for outputting the name/ID number of the office supply and the number to buy;

Example 1:

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while not end of inventory file
  read supply data record from file
  if remaining quantity < minimum quantity
    then
      numbertobuy = maximum quantity - remaining quantity
      output ("Reorder ", ID, name, numbertobuy)
    end if
  end loop
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Example 2:

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for each item in the inventory loop
  with item do
    if remaining quantity < minimum quantity then
      output "Reorder ", ID, name, (maximum quantity - remaining quant
    end if
  end do
end loop
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Examiners report

Candidates were required to write the main steps in an algorithm to output a list of office supplies with the quantity to be ordered. Candidates who gave a list of logical steps written in the style of an algorithm, rather than as a purely descriptive answer, achieved higher marks. Marks were generally lost when candidates failed to recognise the need for a loop to check the whole file, or if they were not specific enough about either what was being ordered, or the quantity of the order. Sometimes, candidates would output the office supply item and its order quantity but failed to show how the order quantity was determined.

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9. [Maximum mark: 6]

An airline has a server that holds the flight database. Passengers can check in using a number of self-service client kiosks located in the airport.

(a.i) Define the term *client*.

[1]

Markscheme

Award [1 max].

A client is a simple application or a whole system that accesses services being provided by a (remote) computer/server;

Examiners report

A large number of candidates recognised a *client* as a system that accesses the services of a *server*.

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(a.ii) Define the term *server*.

[1]

Markscheme

Award [1 max].

A server is a computer / software which manages access to a centralized resource or service in a network.

A server is a computer/software that provides data to other computers/clients in a network;

Examiners report

A large number of candidates recognised a *server* as a computer that provides data to a *client*.

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(b) Explain the functions performed by the server in this situation.

[4]

Markscheme

Award [4 max].

the central computer receives requests from the kiosks/terminals at the check-in desks/mobile phones;
can manage several clients simultaneously (in order, each providing a different set of services);
makes sure that the request is valid by accessing its database;
processes the request;
sends information to the kiosk;
all requests/information are sent over the network connection;

Examiners report

Many candidates failed to understand that answers were required to explain the functions of the server in an airline's system that holds its flight database. Some candidates recognised that the server receives requests from clients, processes requests, accesses the database, or works over the network. However, most candidates only described one or two of these points and rarely stated where the requests came from i.e. the kiosks, check-in desks, or mobile phones.

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10. [Maximum mark: 3]
- (a) Outline **one** feature of a word processor that could reduce the amount of typing required when writing letters. [2]

Markscheme
<i>Award [2 max].</i> Auto-correct/short sequences/codes; can be used to represent longer phrases; Mail merge; allows production of many letters by only typing the (body of a) letter once; Voice recognition; allows easy entry of text;
Examiners report
Candidates were generally able to name or describe a feature of a word processor that could reduce the amount of typing required when writing

letters. Candidates who named a feature and described it achieved both marks.

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- (b) State the purpose of technical documentation provided with software.

[1]

Markscheme

Award [1 max].

Technical documentation explains how to install software;

Technical documentation describes the hardware configuration/operating system needed(to install this software);

Examiners report

A small number of candidates recognised the role of technical documentation in the installation of software or in identifying the hardware requirements of a program. However, many answers were seen in which candidates described user documentation.

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11. [Maximum mark: 1]

Identify **one** common feature found in the user interface of application software to improve its usability.

[1]

Markscheme

Award [1 max]

Toolbars;

Menus;

Dialogue boxes;

Examiners report

Candidates seemed to misunderstand what was required as a feature of the user interface for application software, which would be, for example, a menu or a toolbar, and gave answers such as colour, or answers related to accessibility, such as large fonts.

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12. [Maximum mark: 4]

- (a) Identify **two** reasons why patches may be necessary for an operating system.

[2]

Markscheme

Award [2 max]

Allows bugs/error in operating system to be repaired;

Allows new features to be added to operating system (such as security updates, improving functionality, improving usability, etc.);

Allows compatibility issues to be improved;

Examiners report

Most candidates were able to identify at least one reason why patches may be necessary for an operating system, with a large proportion correctly naming two reasons.

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(b) Identify **two** methods that can be used to obtain these patches.

[2]

Markscheme

Award [2 max]

Automatic patches/updates sent (via internet);

User requested updates (via internet);

Patches sent on CD/DVD/memory stick;

Examiners report

Some candidates correctly identified one or two methods of obtaining patches, for example, by downloading them or installing them from an external device. However, many candidates incorrectly gave the same method twice, for example, by suggesting two methods of downloading patches.

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13. [Maximum mark: 2]
Identify **two** roles that a computer can perform in a network. [2]

Markscheme
<i>Award [2 max]</i> Client; Server/email server/DNS server/file server; Router; Firewall;
Examiners report
Candidates were generally able to identify roles that a computer can perform in a network. However, some candidates incorrectly described actions that a computer performs in a network rather than stating actual roles.

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14. [Maximum mark: 2]

Describe **one** method of implementation for a new computer system.

[2]

Markscheme
<i>Award [2 max]</i> Parallel; old system and new system are operated at the same time until the current system is proved to be successful; Pilot; the new (whole) system is operated in one branch/part of the organization before it is rolled out to the whole organization; Direct; the new system replaces the old system in an immediate switchover; Phased; the new system in phases / stages, gradually replaces parts of the old system until the current system is completely replaced by the new system;
Examiners report
The vast majority of candidates were able to name a method of implementation for a new computer system and then went on to describe it, to achieve both marks. Some candidates, however, successfully named a method, but the description given could have applied to any method of implementation, so it was not specific enough for the second mark.

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15. [Maximum mark: 7]

A group of programmers are involved in creating a new software product. They create many new sub-programs but also use existing sub-programs within the product.

- (a) Outline why a sub-program is considered an example of abstraction.

[2]

Markscheme
<i>Award [2 max]</i> A sub-program is a named section of code that performs a specific task (in a program) / can be called by name / referred by the identifier when needed; without knowing the details (of code and data structures) as these are wrapped / hidden within the sub-program;

Examiners report
Many candidates provided vague or too general answers.

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- (b) Evaluate the use of designing and developing different parts of software products concurrently.

[3]

Markscheme
<i>Award [3 max]</i> different stages (of programming) (<i>Accept examples!</i>) run simultaneously (rather than consecutively);

this decreases product development time / decreases the time to market;
leading to improved productivity/reduces costs;
however, it requires more resources/more software developers;

Examiners report

Many candidates correctly answered this part.

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- (c) Outline **one** way in which users can be informed of software updates.

[2]

Markscheme

Award [2 max]

A message can be sent to the user (When the software is installed and registered, the user provides an email address / phone number);
With a link to the update;

notifications/alerts are sent to the computer (a cookie is placed on the user's computer which communicates with the software developer);
about automatic updates;

(when the program is run) it queries a URL the program developer has built in to check whether the current version matches the latest version;
if not, notifications/alerts are sent;

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16. [Maximum mark: 2]
State **two** compatibility issues that may arise when international businesses merge. [2]

Markscheme
<i>Award [2 max].</i> Language differences / lexical differences present across datasets to be merged; Data representation differences / different data structures (e.g., date format, incompatible file formats); Incompatible hardware; Incompatible operating systems / different software versions;
Examiners report
Candidates were generally able to state at least one compatibility issue related to the merging of international business systems. Some candidates lost out by naming two issues that were too similar, so therefore covering the same marking point.

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17. [Maximum mark: 2]
Outline **one** reason why accurate user documentation is necessary for a system. [2]

Markscheme
<i>Award [2 max].</i> User efficiency; To ensure that users know how to use the system correctly; Support/Troubleshoot; To provide users help when they encounter errors; Accuracy; To ensure the correct methods are used to enable reliable output; Improved user experience; the user is aware of all available features, so they can make the most out of the system;
Examiners report
Candidates were able to identify a reason why accurate user documentation is necessary for a system. A few candidates mixed this up

with system documentation. Some candidates also named two reasons rather than expanding on the one reason, as required by the question.

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